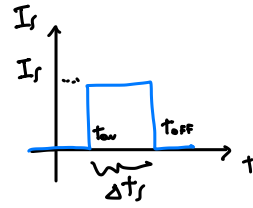
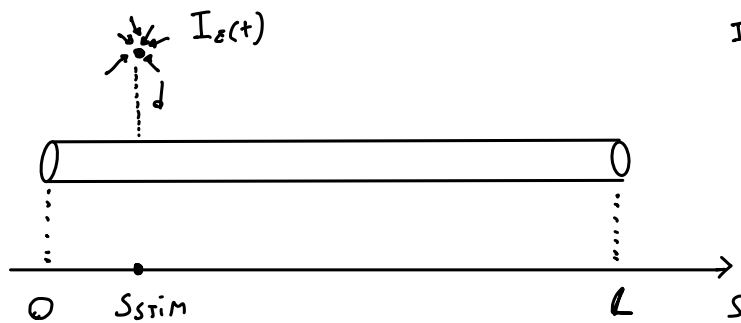


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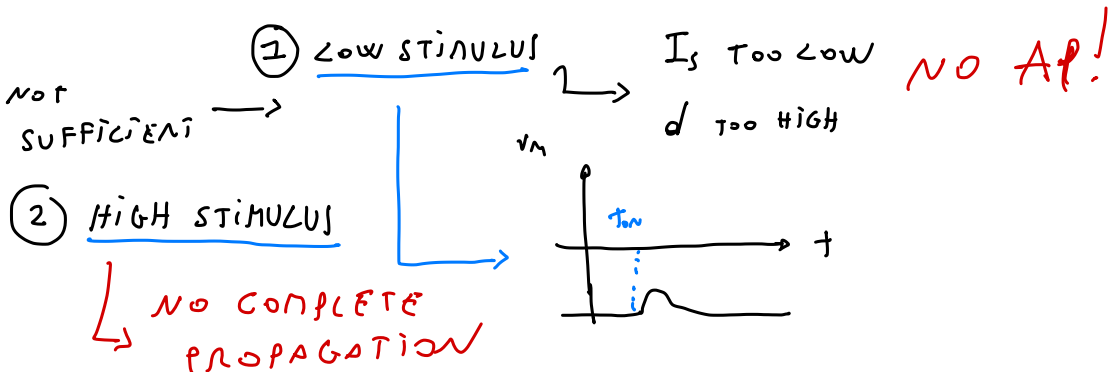
$V_m(s, t)$

$$\frac{dV_m}{dt} = \frac{1}{C_1} \left(\kappa_2 \frac{\partial V_m}{\partial s^2} + \kappa_1 \frac{\partial^2 \phi}{\partial s^2} - I_{ion} \right) \rightarrow \text{FINITE DIFF.}$$

$\hookrightarrow F \rightarrow$ ACTIVATION FUNCTION

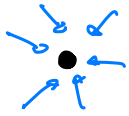
$F > 0$ NECESSARY CONDITION $\leadsto t < t_{on} \begin{cases} V_m = V_h \\ \frac{dV_m}{ds^2} = 0 \\ I_{ion} = 0 \end{cases}$

$t = t_{on} \rightarrow F > 0 \Rightarrow V_m \nearrow$ RECOVER $\frac{dV_m}{dt} > 0$ (DEPOLARIZATION)



$$F(s, t)$$

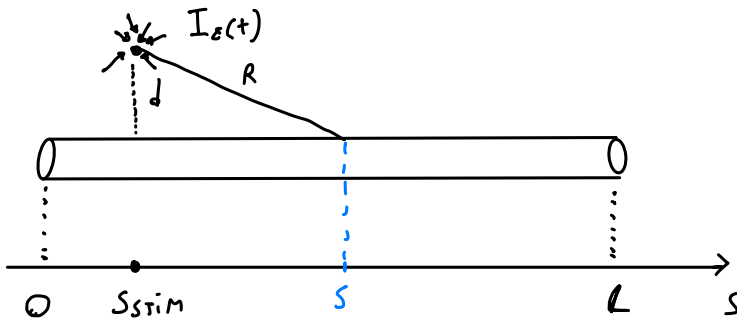
$$t \rightarrow F \neq 0 \quad t_{on} < t < t_{off}$$



$$I_s \text{ WHEN } t_{on} < t < t_{off}$$

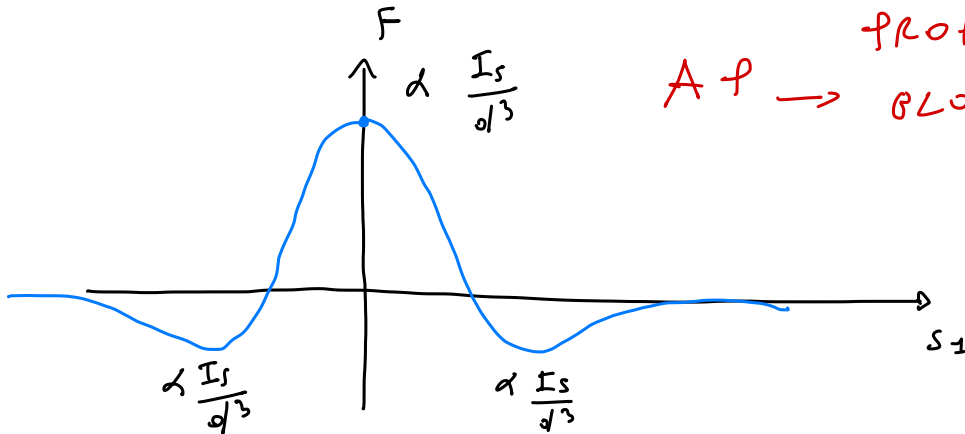
$$\phi_0 = \frac{1}{4\pi G_0} \frac{I_s}{R}$$

$$R = \sqrt{d^2 + (s - s_{stim})^2}$$



$$F = \frac{d^2 \phi_0}{ds^2} = \frac{I_s}{4\pi G_0} \frac{(d^2 - 2s_1^2)}{(d^2 + s_1^2)^{5/2}}$$

$$s_1 = (s - s_{stim})$$



propagation
A P → BLOCKED